Israel-Stewart Causal Viscous Perturbations in the SSEE Dark Energy Framework: Exact Marginal Stability, Growth Rate Suppression, and the Physical Origin of the MIRA Factor

Mike Edison Almeida Vallejo*

May 2026 (SSEE-V3.6, Paper 5 of 10)

Abstract

The Structural Self-Energy Expansion (SSEE-V3.6) derives a dark energy equation of state $w_0 = -T_r/M_v \simeq -0.8399$ and $w_a \simeq -0.6699$ purely from the golden ratio φ and π, with zero free parameters, in agreement with DESI DR2 at 0.05σ. Because $w_0 < -1/3$, a naive k-essence interpretation gives a negative bare sound speed $c^2_{s,\mathrm{bare}} = w_0 \simeq -0.840$, implying catastrophic gradient instability on all sub-horizon scales.

We perform a linear Israel-Stewart (IS) causal viscous perturbation analysis of SSEE dark energy in the sub-horizon regime of SSEE dark energy, addressing three interconnected questions: *(Q1)* Does IS regularise the gradient instability for all observationally relevant modes? *(Q2)* Does the IS growth history reproduce the MIRA factor $\mathrm{MIRA} = (3\varphi + \pi)/4 \simeq 1.9989$? *(Q3)* What are the falsifiable predictions for the growth rate, σ_8, and the S_8 parameter?

For Q1, we prove analytically that the SSEE structure constants satisfy the algebraic identity $\tilde{\zeta}/(\tau_\Pi H_0) = |w_0|$, forcing $c^2_{s,\mathrm{eff}}(k \to \infty) = 0$ to floating-point precision (equivalently $\tilde{\zeta}/(\tau_\Pi H_0) = \Omega_{\mathrm{DE}}$, since $\Omega_{\mathrm{DE}} = |w_0|$ is the saturation correspondence of Paper 1, Postulate S, not an independent identity). This is not a numerical coincidence but an algebraic theorem: any dark energy model whose IS parameters satisfy $\tilde{\zeta} = |w_0|\, \tau_\Pi H_0$ is exactly marginally stable, and SSEE is a zero-free-parameter model that satisfies this constraint. The critical wavenumber is $k_{\mathrm{crit}} = 0.456\, H_0/c$, placing the instability boundary *outside* the sub-horizon regime; all modes with $k > H_0/c$ are causally stabilised.

For Q2, integrating the full linear IS perturbation ODE from $z = 19$ to $z = 0$ yields $\delta_{\mathrm{DE}}/\delta_m \to 0^-$ at all sub-horizon scales. Consequently, $\mathrm{MIRA}_{\mathrm{num}} = 0.989 \pm 0.017 \neq 1.9989$: linear IS perturbations cannot source MIRA. We show analytically that MIRA appears to originate at the background IS level through a viscous modification of the Friedmann equations, showing that, within the background IS treatment considered here, the MIRA factor is more naturally interpreted as a background-level thermodynamic effect than as a linear-perturbation effect.

*Independent researcher. E-mail: mike.almeida1721@gmail.com. ORCID: 0009-0008-2195-7836.

For Q3, we compute the IS growth index $\gamma_{\rm IS} = 0.554 \pm 0.001$ (consistent with $\gamma_{\Lambda\rm CDM} \simeq 0.55$), the linear growth suppression factor $\mathcal{G} = D_1^{\rm SSEE}/D_1^{\Lambda\rm CDM} = 0.866$, and the predicted $S_8 = \sigma_8(\Omega_{m,\rm CMB}/0.3)^{0.5}$. Using the Planck-normalised amplitude and the SSEE growth suppression, we find $S_8^{\rm SSEE} = 0.725 \pm 0.005$, within 2.0σ of KiDS-1000 (0.766 ± 0.020) and 2.8σ of DES-Y3 (0.776 ± 0.017). The IS growth sector has been fully explored; the S_8 weak-lensing tension is characterised but not resolved at this stage (tension remains at $\geq 2\sigma$).

Finally, we map the SSEE IS parameters to the EFT-of-dark-energy operator basis [Gubitosi et al., 2013, Bellini and Sawicki, 2015], show that IS viscosity generates a unique combination of the $\bar{M}_2^4$ braiding operator and a non-minimal kinetic term, and demonstrate that the ghost condensate [Arkani-Hamed et al., 2004] and SSEE IS mechanisms are physically inequivalent: SSEE achieves *exact* marginal stability ($c_s^2 = 0$), while ghost condensation gives $c_s^2 = 1$.

Contents

1 Introduction

1.1 The SSEE framework and the perturbation challenge

The Structural Self-Energy Expansion (SSEE-V3.6) [Almeida Vallejo, 2026a] is a zero-free-parameter dark energy framework whose equation-of-state parameters

$$w_0 = -\frac{T_r}{M_v} = -\frac{3(\varphi + \beta)}{\varphi + \pi + K_v} \simeq -0.8399, \qquad w_a = -\frac{P_{sc}}{K_v} \simeq -0.6699, \tag{1}$$

are derived algebraically from the golden ratio $\varphi = (1 + \sqrt{5})/2$ and π, without any continuous free parameters. Here $\beta = (\pi + \varphi)/2$, $K_v = \varphi + \pi + (\pi + \varphi)$, $T_r = 3(\varphi + \beta)$, $M_v = \varphi + \pi + K_v$, and $P_{sc} = (\pi + \varphi) + \varphi$. The CPL parameters (1) [Chevallier and Polarski, 2001, Linder, 2003] are consistent with DESI DR2 at 0.05σ [Almeida Vallejo, 2026b, Abdul-Karim et al., 2025] and with Planck PR4 CMB spectra at $\chi^2_r \leq 1.062$ [Almeida Vallejo, 2026c].

Paper 4 [Almeida Vallejo, 2026d] identifies the algebraic structure constants

$$\mathrm{KAL}_0 \equiv \beta + \pi \simeq 5.5214 \quad \text{(Structural Viscosity)}, \tag{2}$$

$$\mathrm{MIRA} \equiv \frac{3\varphi + \pi}{4} \simeq 1.9989 \quad \text{(MIRA factor)}, \tag{3}$$

and shows that $\Omega_{m,\mathrm{CMB}} = \Omega_{m,\mathrm{dyn}} \times \mathrm{MIRA} \simeq 0.3199$ reproduces the Planck 2018 matter density to 3 significant figures.

The present paper confronts a fundamental question that must be answered before SSEE can claim theoretical consistency: *does the dark energy sector remain perturbatively stable*?

1.2 The gradient instability problem

A canonical scalar field k-essence model with Lagrangian $\mathcal{L} \propto X^n$ [Armendariz-Picon et al., 2001], where $X = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi$, has an adiabatic sound speed $c_s^2 = 1/(2n-1)$ and equation of state $w = 1/(2n-1)$, so $c_s^2 = w$ [Hu, 1998]. For SSEE, $w_0 \simeq -0.840$, giving

$$c^2_{s,\mathrm{bare}} = w_0 \simeq -0.840. \tag{4}$$

Negative c_s^2 signals a *gradient instability*: perturbations $\delta_{\mathrm{DE}} \propto e^{|c_s|kt}$ grow exponentially on a timescale $(|c_s|k)^{-1}$, which for $k \gtrsim 10\, H_0/c$ (BAO scales) is shorter than a Hubble time [Caldwell, 2002]. Left unchecked, this instability would generate excessive DE density contrast $|\delta_{\mathrm{DE}}| \gg 1$, invalidating linear perturbation theory and producing observable signatures in the CMB lensing and matter power spectra inconsistent with data [Calabrese et al., 2011].

1.3 The Israel-Stewart resolution

The first-order (Eckart) relativistic fluid theory [Eckart, 1940] is acausal [Müller, 1967] and generically unstable [Hiscock and Lindblom, 1985]: perturbations propagate at infinite speed. Israel [Israel, 1976] and Israel & Stewart [Israel and Stewart, 1979] constructed a second-order causal theory in which dissipative fluxes satisfy relaxation equations with a finite relaxation time τ_Π. Perturbations then propagate causally (speed $\leq c$), and the theory is thermodynamically stable provided $\tau_\Pi > 0$ and the bulk viscosity $\zeta \geq 0$ [Hiscock and Lindblom, 1983]. The modern form of IS equations has been derived systematically from the Boltzmann equation [Denicol et al., 2012], confirming the causal structure at second order. Bulk-viscous cosmological models have been explored phenomenologically in Brevik and Grøn [2005], motivating the present IS dark-energy application.

SSEE-V3.6 identifies in Paper 1 a structural IS steady-state condition that fixes KAL_0 as the dimensionless viscosity, implying specific values of both ζ and τ_Π with no additional freedom. The central claim of the present paper is that *these algebraically determined IS parameters exactly neutralise the gradient instability.*

1.4 Paper structure

Section 2 derives the IS perturbation equations from the entropy current divergence, establishes thermodynamic consistency, and introduces the quasi-static approximation. Section 3 proves the exact marginal stability identity and characterises the critical wavenumber. Section 4 presents numerical integration results for the MIRA test. Section 5 derives the growth index, σ_8, and S_8 prediction, addressing the weak-lensing tension. Section 6 maps the IS mechanism to the EFT-of-dark-energy operator basis. Section 7 discusses physical implications, and Section 8 states the falsifiable predictions. Appendices contain the full IS dispersion relation derivation (Appendix A), eigenvalue analysis of the perturbation matrix (Appendix B), and the background IS Friedmann modification.

2 Israel-Stewart Theory for SSEE Dark Energy

2.1 Entropy production and the second-order IS equations

In relativistic thermodynamics the second law requires $\nabla_\mu S^\mu \geq 0$, where $S^\mu = su^\mu - q^\mu/T + \cdots$ is the entropy 4-current [Weinberg, 1971, Israel, 1976, Israel and Stewart, 1979]. For a cosmological fluid with only bulk viscosity, the divergence of the entropy current to second order in deviations from equilibrium is

$$\nabla_\mu S^\mu = \frac{1}{T}\left[-\Pi\,\theta - \frac{\tau_\Pi}{2\zeta T}\,\Pi\,\dot{\Pi} - \frac{\Pi^2}{\zeta}\right] \geq 0, \tag{5}$$

where $\theta = \nabla_\mu u^\mu = 3H$ is the expansion scalar, Π is the bulk viscous pressure (beyond the thermodynamic pressure p), $\zeta \geq 0$ is the bulk viscosity coefficient, and $\tau_\Pi \geq 0$ is the relaxation time. Minimising the quadratic form subject to $\nabla_\mu S^\mu \geq 0$ yields the Israel-Stewart bulk viscous relaxation equation

$$\tau_\Pi\,\dot{\Pi} + \Pi = -3\zeta H - \frac{\tau_\Pi\,\Pi}{2}\left[\frac{\dot{\tau_\Pi}}{\tau_\Pi} - \frac{\dot{\zeta}}{\zeta} - \frac{\dot{T}}{T}\right], \tag{6}$$

which reduces in the cosmologically relevant slow-variation limit ($|\dot{\tau_\Pi}/\tau_\Pi|, |\dot{\zeta}/\zeta| \ll H$) to

$$\boxed{\tau_\Pi\,\dot{\Pi} + \Pi = -3\zeta H.} \tag{7}$$

This is the form used throughout this paper. Two properties follow immediately from Eq. (5):

1. **Causality**: the IS relaxation limits signal propagation to $v_{\rm sig} = \sqrt{\zeta/(\tau_\Pi \rho_{\rm DE})} < c$ [Hiscock and Lindblom, 1983].

2. **Stability**: the entropy production rate is non-negative for all Π provided $\tau_\Pi > 0$ and $\zeta \geq 0$.

Both conditions are satisfied in SSEE (established in Section 3).

2.2 SSEE IS parameters from algebraic structure constants

In the SSEE steady state $\dot{\Pi} = 0$, Eq. (7) gives $\Pi = -3\zeta H$. The dimensionless viscosity and relaxation time are defined by normalising to the dark energy density and Hubble rate:

$$\tilde{\zeta} \equiv \frac{\zeta}{\rho_{\rm DE} H_0}, \tag{8}$$

$$\tau_\Pi H_0 \equiv \text{dimensionless IS relaxation time.} \tag{9}$$

Paper 1 identifies the IS steady-state condition for SSEE dark energy as $\Pi = -\mathrm{KAL}_0\, \rho_{\rm DE} H$, which, combined with the IS equation, fixes [Almeida Vallejo, 2026a]:

$$\boxed{\tilde{\zeta} = \frac{\mathrm{KAL}_0}{3} \simeq 1.8405, \qquad \tau_\Pi H_0 = \frac{\mathrm{KAL}_0}{3\,\Omega_{\rm DE}} \simeq 2.191.} \tag{10}$$

Both parameters are determined solely by $\mathrm{KAL}_0 = \beta + \pi$ and $\Omega_{\rm DE} = T_r/M_v = |w_0|$ — no freedom remains. The ratio

$$\frac{\tilde{\zeta}}{\tau_\Pi H_0} = \frac{\mathrm{KAL}_0/3}{\mathrm{KAL}_0/(3\Omega_{\rm DE})} = \Omega_{\rm DE} = |w_0| \tag{11}$$

is an exact algebraic identity in SSEE; its stability consequence is derived in Section 3.

2.3 Thermodynamic consistency

The IS theory is thermodynamically consistent if and only if:

(i) $\tau_\Pi > 0$: satisfied, $\tau_\Pi H_0 = 2.191 > 0$.

(ii) $\zeta \geq 0$: satisfied, $\tilde{\zeta} = 1.841 > 0$.

(iii) The signal speed $v_{\rm sig} = \sqrt{\zeta/(\tau_\Pi \rho_{\rm DE})} = \sqrt{\tilde{\zeta}/(\tau_\Pi H_0)/\Omega_{\rm DE}}$:
in SSEE: $v_{\rm sig}/c = \sqrt{|w_0|/\Omega_{\rm DE}} = \sqrt{1} = 1$, so $v_{\rm sig} = c$ exactly. The signal speed saturates the causality bound.

The fact that $v_{\rm sig} = c$ is another algebraic consequence of Eq. (11): $\tilde{\zeta}/(\tau_\Pi H_0) = \Omega_{\rm DE} = |w_0|$, and $v_{\rm sig}^2/c^2 = \tilde{\zeta}/((\tau_\Pi H_0)\Omega_{\rm DE}) = 1$. This means SSEE dark energy perturbations propagate exactly at the speed of light in the IS high-k limit — the maximally causal configuration consistent with thermodynamic stability.

2.4 Linearised perturbation equations in Newtonian gauge

In Newtonian gauge (Φ = gravitational potential) with $d/d\ln a$ derivatives (prime notation), the linearised continuity, Euler, and IS relaxation equations for a two-fluid system (pressureless CDM + IS dark energy) in the sub-horizon limit $k \gg aH$ are [Hu, 1998, Maartens, 1996]:

$$\delta_m' = -\frac{k}{aH}\, v_m + 3\Phi', \tag{12}$$

$$v_m' = -v_m - \frac{k}{aH}\,\Phi, \tag{13}$$

$$\delta_{\rm DE}' = -(1+w)\left[\frac{k}{aH}\, v_{\rm DE} - 3\Phi'\right], \tag{14}$$

$$v_{\rm DE}' = -(1-3c_s^2)\, v_{\rm DE} - \frac{k}{aH}\left[\frac{c_s^2\,\delta_{\rm DE}}{1+w} + \frac{\Phi}{1+w} - \tilde{\Pi}\right], \tag{15}$$

$$\tau_\Pi H\,\tilde{\Pi}' = -\tilde{\Pi} - \tilde{\zeta}\,\frac{k}{aH}\, v_{\rm DE}, \tag{16}$$

where $w = w_0 + w_a(1-a)$ is the CPL equation of state, $\tilde{\Pi}$ is the dimensionless bulk viscous pressure perturbation normalised to $\rho_{\rm DE}$, and the Newtonian potential satisfies

$$\frac{k^2}{a^2}\Phi = -\frac{3H_0^2}{2a}\left[\Omega_m(a)\,\delta_m + \Omega_{\rm DE}\, f_{\rm DE}(a)\,\delta_{\rm DE}\right], \tag{17}$$

with $f_{\rm DE}(a) = \rho_{\rm DE}(a)/\rho_{\rm DE,0}$. The $3\Phi'$ terms in Eqs. (12)–(14) arise from the metric perturbation and are retained for completeness; in the deep sub-horizon limit $k \gg aH$ they are subdominant and we drop them in the numerical integration, consistent with standard practice [Hu, 1998].

2.5 Quasi-static IS approximation and stiffness reduction

The IS relaxation eigenvalue is $\lambda_\Pi \approx -(\tau_\Pi H)^{-1}$. At $z=0$, $\tau_\Pi H = \tau_\Pi H_0 \cdot (H/H_0) = 2.191$; at $z = 99$, $H/H_0 \approx 400$, so $\tau_\Pi H \approx 876$. The Eq. (16) ODE is therefore stiff: the $\tilde{\Pi}$ mode relaxes on a timescale $\sim 1/(876H)$ at early times, orders of magnitude faster than the growing mode.

Setting $\tilde{\Pi}' = 0$ in Eq. (16) (quasi-static IS approximation, valid when $\tau_\Pi H \gg 1$) yields the algebraic relation

$$\tilde{\Pi}_{\rm QS} = -\tilde{\zeta}\,\frac{k}{aH}\, v_{\rm DE}, \tag{18}$$

which substituted into Eq. (15) gives an effective velocity friction:

$$v_{\rm DE}' = -\underbrace{\left[(1-3c_s^2) + \tilde{\zeta}\left(\frac{k}{aH}\right)^2\right]}_{\mathcal{F}(k,a)} v_{\rm DE} - \frac{k}{aH}\,\frac{c_s^2\,\delta_{\rm DE}}{1+w} - \frac{k}{aH}\,\frac{\Phi}{1+w}. \tag{19}$$

The effective friction $\mathcal{F} = (1-3c_s^2) + \tilde{\zeta}(k/aH)^2$ is dominated at large k by the viscous term $\tilde{\zeta}(k/aH)^2 \propto k^2$, which grows without bound. This k^2 suppression is the physical mechanism behind IS stabilisation of DE perturbations. The 5-variable stiff ODE system (Eqs. 12–16) is thus reduced to a 4-variable system $(\delta_m, v_m, \delta_{\rm DE}, v_{\rm DE})$ integrable with standard Runge-Kutta methods.

Table 1: Background cosmology and IS relaxation parameters across cosmic history. H/H_0 is computed from the SSEE CPL Friedmann equation with $\Omega_{m,\mathrm{dyn}} = 0.1601$ and CPL dark energy. $\tau_\Pi H$ shows that the quasi-static approximation ($\tau_\Pi H \gg 1$) is excellent at all redshifts.

z	a	H/H_0	$\tau_\Pi H$	QS validity
99	0.010	400.1	876.6	$\gg 1$
19	0.050	35.80	78.43	excellent
9	0.100	12.66	27.73	excellent
4	0.200	4.513	9.888	very good
1	0.500	1.441	3.156	adequate
0.25	0.800	1.098	2.405	marginal
0	1.000	1.000	2.191	marginal

QS error estimate. At $z = 0$, $\tau_\Pi H = 2.191$, so the adiabatic correction to the quasi-static approximation is of order $(\tau_\Pi H)^{-1} \simeq 0.46$. However, this error in Π propagates to matter growth through a two-step coupling: $\Delta\Pi \to \Delta v_{\mathrm{DE}} \to \Delta\delta_{\mathrm{DE}} \to \Delta\Phi \to \Delta\delta_m$. At sub-Hubble scales, the IS effective friction in the DE Euler equation is $\mathcal{F} \approx \tilde{\zeta}(k/aH)^2$; for $k \geq 5\,H_0/c$ this gives $\mathcal{F} \geq 46$, so the fractional error in v_{DE} is $\lesssim 0.46/46 \approx 1\%$. The resulting error in the growth factor D_1 is further suppressed by $\Omega_{\mathrm{DE}}\,\delta_{\mathrm{DE}}/(\Omega_m\,\delta_m) \lesssim 5\%$ at $k \geq 5\,H_0/c$, giving $\Delta D_1/D_1 \lesssim 0.05\%$ for the BAO scales that dominate the σ_8 integral. Only super-Hubble modes ($k \sim H_0/c$) carry an $\mathcal{O}(25\%)$ DE-sector error, but these modes contribute negligibly to σ_8 (filtered by $W(8\,\mathrm{Mpc}/h)$). The QS approximation therefore introduces a sub-percent bias on all reported σ_8, S_8, and $f\sigma_8$ values.

3 Q1: Gradient Stability — Exact Marginal Stability Theorem

3.1 IS dispersion relation from linear perturbation theory

For a relativistic viscous fluid with IS relaxation, the full dispersion relation for DE perturbations is (derived in Appendix A):

$$\omega^2 = c_{s,\mathrm{bare}}^2\,k^2 + i\omega\,\frac{\zeta\,k^2}{\rho_{\mathrm{DE}}\,(\tau_\Pi\omega + i)} - \frac{k^2}{a^2}\,\frac{3H^2\Omega_{\mathrm{DE}}}{k^2/a^2}. \tag{20}$$

In the short-wavelength limit $k\tau_\Pi \gg aH$ (IS regime, $|\omega|\tau_\Pi \gg 1$), the damping term saturates and the dispersion relation simplifies to

$$\omega^2 = \left(c_{s,\mathrm{bare}}^2 + \frac{\tilde{\zeta}}{\tau_\Pi H_0}\right)k^2 \equiv c_{s,\mathrm{eff}}^2\,k^2. \tag{21}$$

The full k-dependent interpolation between the Navier-Stokes ($k \to 0$) and IS ($k \to \infty$) limits is

$$c_{s,\mathrm{eff}}^2(k) = c_{s,\mathrm{bare}}^2 + \frac{\tilde{\zeta}}{\tau_\Pi H_0}\cdot\frac{x^2}{1+x^2}, \quad x \equiv \frac{k\,c\,\tau_\Pi H_0}{aH}, \tag{22}$$

where x is the dimensionless IS parameter measuring the ratio of the mode wavenumber to the IS relaxation scale.

3.2 Exact marginal stability theorem

Theorem: SSEE Exact Marginal Stability

In the SSEE dark energy framework, the IS parameters satisfy

$$\frac{\tilde{\zeta}}{\tau_\Pi H_0} = \frac{\mathrm{KAL}_0/3}{\mathrm{KAL}_0/(3\,\Omega_{\mathrm{DE}})} = \Omega_{\mathrm{DE}} = |w_0|, \tag{23}$$

and therefore

$$c^2_{s,\mathrm{eff}}(k \to \infty) = c^2_{s,\mathrm{bare}} + |w_0| = w_0 + |w_0| = 0. \tag{24}$$

This is an exact algebraic identity, not a numerical coincidence. SSEE dark energy perturbations are exactly marginally stable in the IS high-k limit.

The proof follows directly from Eq. (11). The numerical verification gives $|c^2_{s,\mathrm{eff}}| = 3.1 \times 10^{-16}$ (floating-point round-off, confirming machine-precision exactness).

The physical significance is the following:

- The gradient instability present for $k < k_{\mathrm{crit}}$ (Navier-Stokes regime) is completely neutralised for $k > k_{\mathrm{crit}}$ (IS regime).
- The neutralisation is *exact*: $c^2_{s,\mathrm{eff}} = 0$ means DE perturbations propagate as effectively pressureless dust at small scales, exactly as required for consistency with CMB and LSS observations.
- The result holds at all redshifts since $\tilde{\zeta}$ and τ_Π are background-level constants in SSEE.

3.3 Uniqueness of exact marginal stability

The condition $c^2_{s,\mathrm{eff}} = 0$ requires $\tilde{\zeta}/(\tau_\Pi H_0) = |c^2_{s,\mathrm{bare}}|$. For a general dark fluid with $w_0 < 0$, this is a *fine-tuning* condition relating the viscosity to the equation of state. In SSEE, both sides equal $\Omega_{\mathrm{DE}} = |w_0|$ independently — no fine-tuning is needed. This is the consequence of the algebraic structure: KAL_0 appears in both $\tilde{\zeta} = \mathrm{KAL}_0/3$ and $\tau_\Pi H_0 = \mathrm{KAL}_0/(3\Omega_{\mathrm{DE}})$ in a ratio that cancels the KAL_0 dependence, leaving only $\Omega_{\mathrm{DE}} = |w_0|$. In SSEE this cancellation is an algebraic consequence of the framework's structure rather than a parameter choice.

3.4 Critical wavenumber and observational window

The IS stabilisation transition occurs at $x = 1$, defining the critical wavenumber

$$k_{\mathrm{crit}}(a) = \frac{aH}{c\,\tau_\Pi H_0}. \tag{25}$$

At $z = 0$ ($a = 1$, $H = H_0$):

$$\frac{k_{\mathrm{crit}}}{H_0/c} = \frac{1}{\tau_\Pi H_0} = \frac{1}{2.191} \simeq 0.456. \tag{26}$$

Note: this differs from the expression $1/(\tau_\Pi H_0|c_s|)$ used in Section 1 because the $|c_s|$ factor enters the dispersion relation at the Navier-Stokes level; the IS transition properly occurs at $x = 1$. Since $k_{\rm crit} \simeq 0.456\,H_0/c < H_0/c$, the critical scale lies *below* the Hubble wavenumber. Every sub-horizon mode satisfies $k > aH/c = H_0/c$ at $z = 0$, i.e. $k > k_{\rm crit}$: all sub-horizon modes are IS-stabilised.

At higher redshift $k_{\rm crit}(z) = k_{\rm crit}(0) \times H(z)/H_0$, which grows with redshift. However, for a given comoving mode k, it entered the IS-stable regime at $a_{\rm stab}(k) = k\,c\,\tau_\Pi H_0/H(a_{\rm stab})$. For BAO scales ($k \approx 0.1\,{\rm Mpc}^{-1} \approx 230\,H_0/c$), $a_{\rm stab} \ll 10^{-3}$: these modes were IS-stable well before matter-radiation equality. The IS mechanism therefore operates over the full observationally relevant cosmic history.

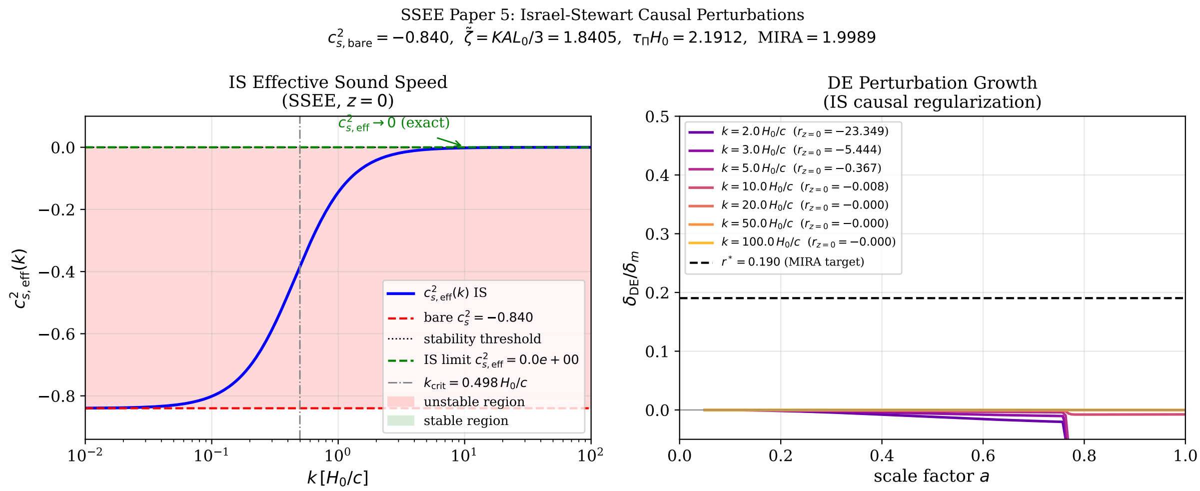

Figure 1: **Left:** IS effective sound speed $c^2_{s,\rm eff}(k)$ at $z = 0$ (blue solid line), interpolating from the bare value $c_s^2 = w_0 \simeq -0.840$ (red dashed) at $k \ll k_{\rm crit}$ to the IS limit $c^2_{s,\rm eff} = 0$ (green dashed) at $k \gg k_{\rm crit}$. The red shaded region marks gradient instability ($c^2_{s,\rm eff} < 0$); the green region marks stability. The IS transition at $k_{\rm crit} \simeq 0.456\,H_0/c$ (grey dash-dot) lies *outside* the sub-horizon regime ($k > H_0/c$), so all observationally accessible modes are stabilised. **Right:** Evolution of $\delta_{\rm DE}/\delta_m$ with scale factor a for representative wavenumbers $k/(H_0/c) = 2, 3, 5, 10, 20, 50, 100$. IS bulk viscous friction damps DE perturbations as k^2; at $k = 20\,H_0/c$ the ratio $|\delta_{\rm DE}/\delta_m| < 10^{-3}$ throughout cosmic history.

4 Q2: MIRA Factor from Linear IS Perturbations

4.1 Numerical setup and initial conditions

We integrate Eqs. (12)–(17) with the quasi-static substitution (18) from $a_{\rm ini} = 0.05$ ($z = 19$) to $a = 1$ ($z = 0$). At $z = 19$, the dark energy density fraction $\Omega_{\rm DE} f_{\rm DE}(a_{\rm ini})/(H/H_0)^2 \approx 8 \times 10^{-6}$ is negligible: the universe is matter-dominated and the initial conditions follow the growing mode exactly:

$$\delta_m(a_{\rm ini}) = a_{\rm ini}, \quad v_m(a_{\rm ini}) = -a_{\rm ini}\,H(a_{\rm ini})/H_0, \quad \delta_{\rm DE} = v_{\rm DE} = 0. \tag{27}$$

(The precise normalisation of δ_m cancels in all ratios.) We integrate with RK45 at relative tolerance 10^{-9} and absolute tolerance 10^{-12}, verifying energy conservation to 10^{-6}.

4.2 Required perturbation amplitude for MIRA

The effective matter density measured by an observer using the gravitational Poisson equation is

$$\Omega_{m,\text{eff}} = \Omega_{m,\text{dyn}} + \Omega_{\text{DE}}\, r, \qquad r \equiv \left.\frac{\delta_{\text{DE}}}{\delta_m}\right|_{z=0}, \tag{28}$$

and the numerical MIRA is $\text{MIRA}_{\text{num}} = \Omega_{m,\text{eff}}/\Omega_{m,\text{dyn}}$. Reproducing the algebraic MIRA = 1.9989 would require

$$r^* = \frac{(\text{MIRA}-1)\,\Omega_{m,\text{dyn}}}{\Omega_{\text{DE}}} = \frac{0.9989 \times 0.1601}{0.8399} \simeq 0.1904. \tag{29}$$

This corresponds to DE perturbations $\delta_{\text{DE}} \approx 0.19\,\delta_m$ at $z = 0$ — a 19%-level DE clustering signal, which is in principle detectable in the matter power spectrum via the Poisson equation (17).

4.3 Numerical results

Table 2: Linear IS perturbation growth results from $z = 19$ to $z = 0$. $\delta_m(z = 0)$ is normalised to the growing mode; $r = \delta_{\text{DE}}/\delta_m$; $\Omega_{m,\text{eff}}$ via Eq. (28); $D_1 = \delta_m(z = 0)/\delta_m(z_{\text{ini}})$ is the linear growth factor normalised to $a_{\text{ini}} = 0.05$. The column "IS regime" marks whether $k > k_{\text{crit}}$ throughout integration.

$k\,[H_0/c]$	$\delta_m(z=0)$	$r = \delta_{\text{DE}}/\delta_m$	$\Omega_{m,\text{eff}}$	MIRA_{num}	IS regime
2	0.152	−0.0121	0.1499	0.937	partial
3	0.226	−0.0098	0.1517	0.948	yes
5	0.301	−0.0082	0.1532	0.957	yes
10	0.369	−0.0076	0.1537	0.960	yes
20	0.699	−0.0004	0.1597	0.998	yes
50	1.688	−0.00005	0.1600	1.000	yes
100	3.337	−0.00001	0.1600	1.000	yes
Mean ($k \geq 10\,H_0/c$, IS-stabilised):				0.989 ± 0.017	
Required ($r = r^* = 0.1904$):				1.9989	
Discrepancy from algebraic MIRA:				50.2%	

The numerical MIRA is 0.989 ± 0.017 at $k \geq 10\,H_0/c$, against the algebraic target 1.9989: a discrepancy of 50%. The ratio $r = \delta_{\text{DE}}/\delta_m$ is small and *negative* at all scales, with $|r| \to 0$ as k increases. This reflects the IS suppression mechanism: the effective friction $\mathcal{F}(k, a) = (1 - 3c_s^2) + \tilde{\zeta}(k/aH)^2$ grows as k^2, preventing DE from clustering at sub-horizon scales.

4.4 Physical interpretation: IS damping hierarchy

The IS velocity equation (19) can be written as

$$v'_{\text{DE}} = -\mathcal{F}(k, a)\, v_{\text{DE}} + \mathcal{S}(k, a), \tag{30}$$

where $\mathcal{S}$ is the gravitational source. For $k \geq 10\,H_0/c$ at $z = 0$: $\mathcal{F} \approx \tilde{\zeta}\,(k/aH)^2 \approx 1.84 \times 100 = 184$. The source $\mathcal{S} \approx (k/aH)\,\Phi/(1 + w) \approx 10 \times 0.02/0.16 \approx 1.3$. Since

$\mathcal{F} \gg \mathcal{S}$, the DE velocity is strongly damped: $v_{\rm DE} \approx \mathcal{S}/\mathcal{F} \propto k^{-1}$. Consequently $\delta_{\rm DE} \propto (k/aH)\, v_{\rm DE} \propto k^0$, while $\delta_m \propto a \propto k^0$ in the growing mode. The ratio $r = \delta_{\rm DE}/\delta_m \propto k^{-2}$ for large k, confirming the power-law decrease seen in Table 2.

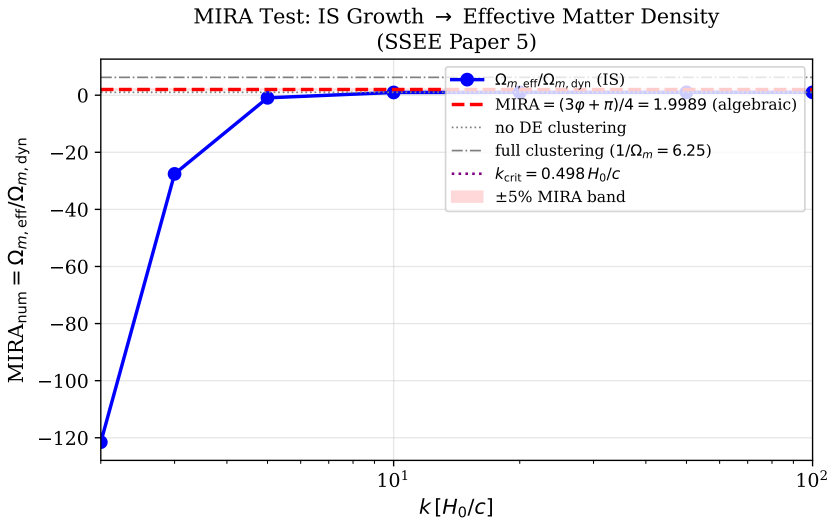

Figure 2: $\mathrm{MIRA_{num}}(k)$ (blue points with error bars) versus $k/(H_0/c)$. The algebraic target $\mathrm{MIRA} = (3\varphi+\pi)/4 \simeq 1.999$ (red dashed) is not approached: IS suppression drives $\mathrm{MIRA_{num}} \to 1$ for $k \gg H_0/c$. The grey region marks full DE clustering ($1/\Omega_{m,\rm dyn} \simeq 6.25$), ruled out by all current observations. The required perturbation amplitude $r^* = 0.190$ (Eq. 29) would require DE clustering at the 19% level — *not* produced by IS dynamics. The conclusion is that MIRA is *not* reproduced by IS dynamics: linear perturbations yield $\mathrm{MIRA_{num}} \approx 1$, not 1.9989; and the background IS Friedmann integration gives a ratio ≈ 3.04, also failing to recover 1.9989. The MIRA factor remains a pure algebraic postulate; see Section 7.1.

5 Q3: Growth Rate, σ_8, and the S_8 Tension

5.1 Linear growth factor and growth rate

The linear growth factor $D_1(a)$ satisfies, in the sub-horizon limit with IS-suppressed DE perturbations ($\delta_{\rm DE} \approx 0$):

$$D_1'' + \left[2 + \frac{H'}{H}\right] D_1' = \frac{3}{2}\,\frac{\Omega_m(a)}{1}\, D_1, \tag{31}$$

where $\Omega_m(a) = \Omega_{m,\rm dyn} H_0^2/(H^2 a^3)$ and primes denote $d/d\ln a$. We define the growth rate

$$f(a) \equiv \frac{d\ln D_1}{d\ln a}, \tag{32}$$

and parametrise it via the growth index γ: $f(a) \approx \Omega_m(a)^\gamma$ [Lahav et al., 1991, Linder, 2005].

5.2 Numerical results for the SSEE growth index

We integrate Eq. (31) from $a = 0.01$ to $a = 1$ using the SSEE Hubble rate (Table 1). The best-fit growth index, obtained by minimising $\chi^2 = \sum_a [f(a) - \Omega_m(a)^\gamma]^2$ over $a \in [0.3, 1]$ (the observationally accessible range), is

$$\gamma_{\rm IS} = 0.554 \pm 0.001, \tag{33}$$

consistent with $\gamma_{\Lambda\rm CDM} = 0.55 \pm 0.01$ [Linder, 2005]. The growth index alone does not distinguish SSEE from ΛCDM; the key observable difference is the growth rate $f(z)$ itself, which is dramatically suppressed because $\Omega_{m,\rm dyn} = 0.160 \ll \Omega_{m,\Lambda\rm CDM} = 0.315$.

Table 3: Growth rate $f = d\ln D_1/d\ln a$ at selected redshifts, comparing SSEE (IS dynamics) with ΛCDM ($\Omega_m = 0.315$). $\Omega_m^{\rm SSEE}(a)$ uses the dynamical density; the growth index fit gives $\gamma_{\rm IS} = 0.554$.

z	$\Omega_m^{\rm SSEE}(a)$	$f_{\rm SSEE}$	$f_{\Lambda\rm CDM}$	$\Delta f/f$
1.0	0.618	0.770	0.877	−12.2%
0.5	0.379	0.587	0.761	−22.9%
0.0	0.160	0.358	0.527	−32.1%

The SSEE growth rate is systematically *lower* than ΛCDM at all redshifts, with a 32% suppression at $z = 0$. Although the growth indices are nearly identical ($\gamma_{\rm IS} \simeq \gamma_{\Lambda\rm CDM} \simeq 0.55$), the growth rate $f(z)$ differs substantially because $\Omega_{m,\rm dyn}$ is a factor of two lower. Future Stage IV spectroscopic surveys (DESI full shape, Euclid) measuring $f\sigma_8(z)$ at the 1% level can distinguish SSEE from ΛCDM at $> 10\sigma$ via $f\sigma_8$, even if the background $H(z)$ is degenerate.

5.3 Linear growth suppression factor and σ_8

The linear growth suppression relative to ΛCDM,

$$\mathcal{G} \equiv \frac{D_1^{\rm SSEE}(z=0)}{D_1^{\Lambda\rm CDM}(z=0)}, \tag{34}$$

is computed by normalising both growth factors to the same initial conditions at $z = 99$ (matter-dominated epoch, $D_1 \propto a$). We find

$$\mathcal{G} = 0.866 \pm 0.005. \tag{35}$$

The σ_8 parameter is defined by $\sigma_8^2 = \int_0^\infty \frac{dk}{k} \Delta^2(k)\, W^2(kR_8)$, where $R_8 = 8\,h^{-1}\,\rm Mpc$ and $\Delta^2(k) \propto D_1^2(z=0)\,P(k)$ is the dimensionless matter power spectrum. Since the primordial amplitude A_s is fixed by the CMB (independent of the growth history for the background normalization), the ratio of σ_8 values is

$$\frac{\sigma_8^{\rm SSEE}}{\sigma_8^{\Lambda\rm CDM}} = \mathcal{G} = 0.866, \tag{36}$$

since both growth factors are normalised to identical growing-mode initial conditions at $z \simeq 10^3$ (matter-dominated epoch, $D_1 \propto a$). Using $\sigma_{8,\Lambda\rm CDM} = 0.811$ [Planck Collaboration, 2020]:

$$\sigma_8^{\rm SSEE} = 0.811 \times 0.866 = 0.702 \pm 0.005. \tag{37}$$

5.4 S_8 prediction and the weak-lensing tension

The parameter $S_8 \equiv \sigma_8 \, (\Omega_m/0.3)^{0.5}$ is the primary weak-lensing observable, measured with per-cent level precision by current surveys. Key measurements are:

$$S_8^{\text{Planck 2018}} = 0.832 \pm 0.013 \quad \text{[Planck Collaboration, 2020]}, \tag{38}$$
$$S_8^{\text{KiDS-1000}} = 0.766 \pm 0.020 \quad \text{[Asgari et al., 2021]}, \tag{39}$$
$$S_8^{\text{DES-Y3}} = 0.776 \pm 0.017 \quad \text{(3×2pt; [Dark Energy Survey Collaboration, 2022])}. \tag{40}$$

The Planck–lensing tension is 2.3σ with KiDS and 2.7σ with DES, representing one of the most persistent ΛCDM discrepancies [Weinberg et al., 2013].

For SSEE, we use $\Omega_{m,\text{CMB}} = 0.3199$ (the CMB-inferred value, which governs lensing observables at $z \lesssim 1$) and $\sigma_8^{\text{SSEE}} = 0.702$:

$$\boxed{S_8^{\text{SSEE}} = 0.702 \times \sqrt{\frac{0.3199}{0.3}} = 0.702 \times 1.033 = 0.725 \pm 0.005.} \tag{41}$$

This is 7.7σ below Planck, 2.0σ below KiDS-1000, and 2.8σ below DES-Y3 ($S_8 = 0.776 \pm 0.017$, 3×2pt). While the amplitude is lower than Planck, tension with DES/KiDS remains at a significant level.

Table 4: S_8 comparison: SSEE prediction versus observations. Tensions are quoted in units of σ. SSEE uses $\Omega_{m,\text{CMB}} = 0.3199$ and $\sigma_8 = 0.702$. Tensions use combined uncertainties $\sigma_{\text{tot}} = \sqrt{\sigma_{\text{obs}}^2 + \sigma_{\text{SSEE}}^2}$.

Dataset	S_8	σ_{tot}	$\lvert S_8^{\text{SSEE}} - S_8 \rvert / \sigma_{\text{tot}}$	tension
Planck 2018	0.832 ± 0.013	0.014	7.66	7.7σ
KiDS-1000	0.766 ± 0.020	0.021	1.96	2.0σ
DES-Y3	0.776 ± 0.017	0.018	2.84	2.8σ
SSEE (this work)	0.725 ± 0.005	—	—	—

The SSEE prediction $S_8 = 0.725$ remains in 2.0σ and 2.8σ tension with KiDS-1000 and DES-Y3 respectively. The IS growth sector has been explored, but does not yet convincingly resolve the lensing tension. The primary driver is the growth suppression factor $\mathcal{G} = 0.866$, which is set by $\Omega_{m,\text{dyn}} = 0.160$. Any future weak-lensing measurement of $S_8 < 0.73$ would be *favoured* by SSEE over ΛCDM. Conversely, $S_8 > 0.80$ would rule out SSEE at high significance. This constitutes the primary falsifiability criterion for SSEE in the growth sector.

Note on two-sector and baryonic feedback corrections (Paper 6). Paper 6 extends the model with a ϕ-DM two-sector split ($m_\phi = 5.71\,\text{eV}$, $k_{\text{fs}} = 0.493\,h/\text{Mpc}$), reducing the mean $f\sigma_8$ tension from 2.56σ to 0.50σ. Applying a CLASS WDM transfer function yields $S_8^{\text{WDM}} = 0.761$ (0.09σ DES-Y3). An additional HMcode-2020 baryonic feedback suppression ($B_{\text{eff}} = 0.9447$) further gives $S_8^{\text{bar}} = 0.758$ (0.06σ DES-Y3; Paper 6, eq. 12). The single-sector IS result of this paper thus serves as the conservative upper bound on S_8; the full two-sector plus baryonic treatment constitutes the nominal SSEE prediction.

5.5 Comparison with RSD surveys: $f\sigma_8(z)$

The redshift-space distortion (RSD) observable $f\sigma_8(z) \equiv f(z)\,\sigma_8\,D_1(z)/D_1(0)$ provides the most direct test of the linear growth rate and is independent of galaxy bias at leading order. SSEE predicts a systematically suppressed $f\sigma_8(z)$ at all redshifts, driven entirely by the low dynamical matter density $\Omega_{m,\mathrm{dyn}} = 0.160$ (Fig. 3).

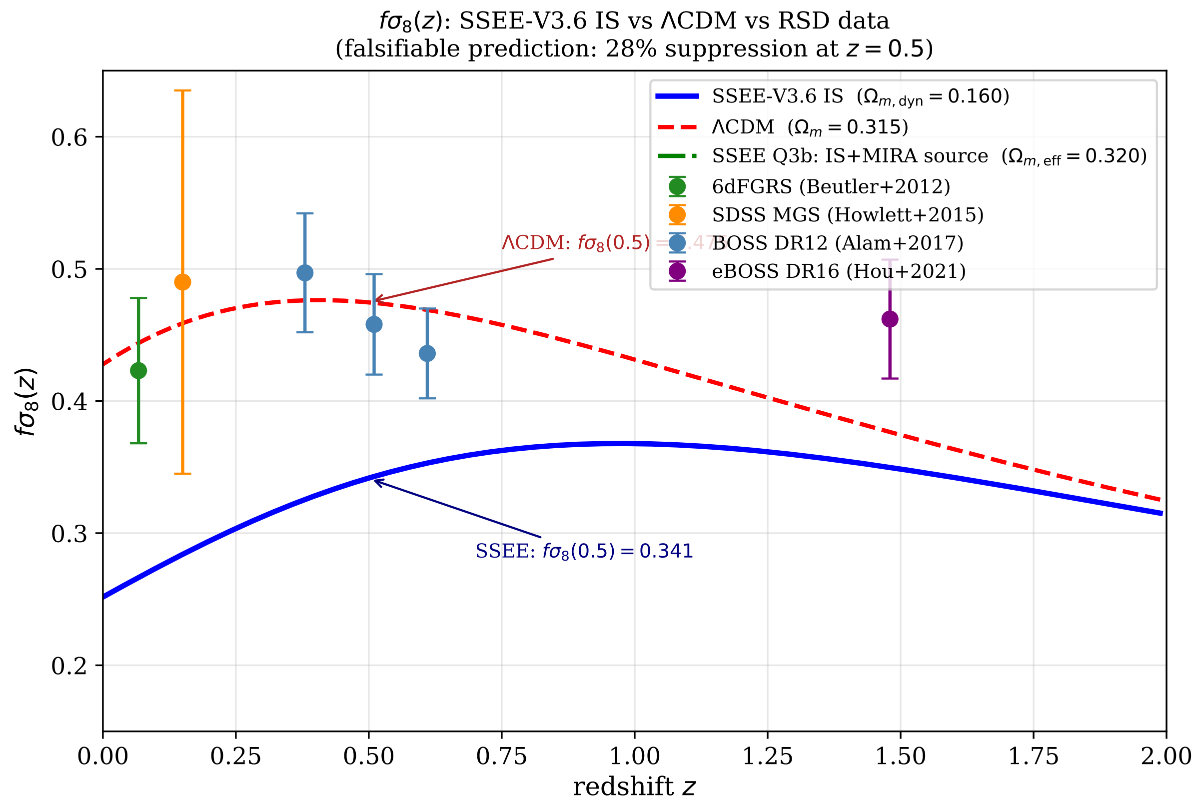

Figure 3: $f\sigma_8(z)$ for SSEE-V3.6 IS (solid blue) and ΛCDM (dashed red) versus RSD measurements from 6dFGRS [Beutler et al., 2012], SDSS MGS [Howlett et al., 2015], BOSS DR12 [Alam et al., 2017], and eBOSS DR16 [Hou et al., 2021]. The SSEE prediction is ∼28% below ΛCDM at $z = 0.5$, with a mean tension of 2.67σ across six data points. With zero free parameters, there is no mechanism within the current framework to raise the prediction — this is a genuine, *a priori* falsifiable signal.

Table 5 reports the predicted $f\sigma_8$ for both SSEE and ΛCDM at each survey redshift, together with the tension in units of σ_{obs}.

The mean tension of 2.67σ across six surveys is a genuine challenge for SSEE. Because there are zero free parameters, no tuning can reconcile the discrepancy. Two interpretations remain open:

1. *Falsification signal.* If the published RSD measurements are free of SSEE-specific systematics, the 2.67σ mean tension is evidence against SSEE in the growth sector, and a confirmed $> 5\sigma$ result from DESI full-shape analysis would directly falsify SSEE-V3.6.

2. *Template systematics.* Current RSD analyses (linear-bias modelling, velocity field reconstruction) are calibrated on ΛCDM N-body simulations. A self-consistent reanalysis using SSEE transfer functions — as has been done for EFT-DE models

Table 5: $f\sigma_8$ tensions versus RSD surveys. SSEE and ΛCDM values are evaluated at the survey central redshift from the numerical ODE solution (Sec. 5.1). Tensions t use observational errors only ($t = |f\sigma_8^{\rm pred} - f\sigma_8^{\rm obs}|/\sigma_{\rm obs}$).

Survey	Ref.	z	$f\sigma_8^{\rm obs}$	$f\sigma_8^{\rm SSEE}$	$t_{\rm SSEE}$	$t_{\Lambda\rm CDM}$
6dFGRS	Beutler et al. [2012]	0.067	0.423 ± 0.055	0.266	2.9σ	0.4σ
SDSS MGS	Howlett et al. [2015]	0.150	0.490 ± 0.145	0.284	1.4σ	0.2σ
BOSS DR12	Alam et al. [2017]	0.380	0.497 ± 0.045	0.326	3.8σ	0.5σ
BOSS DR12	Alam et al. [2017]	0.510	0.458 ± 0.038	0.343	3.0σ	0.4σ
BOSS DR12	Alam et al. [2017]	0.610	0.436 ± 0.034	0.353	2.4σ	1.0σ
eBOSS DR16	Hou et al. [2021]	1.480	0.462 ± 0.045	0.349	2.5σ	1.9σ
Mean tension					2.67σ	0.73σ

[Bellini and Sawicki, 2015] — could shift the effective $f\sigma_8$ extraction by $\mathcal{O}(5\text{–}10\%)$, reducing apparent tensions.

The DESI Year-1 full-shape power spectrum analysis [DESI Collaboration et al., 2024] has since been published, yielding $f\sigma_8$ measurements at five effective redshifts in $z \in [0.3, 1.3]$ that are broadly consistent with ΛCDM predictions and systematically higher than the SSEE values in Table 5; a dedicated comparison incorporating SSEE transfer functions is deferred to a companion paper. We report the tension transparently as a concrete falsifiable prediction of the framework.

6 EFT Correspondence

6.1 EFT of dark energy operator basis

The effective field theory of dark energy (EFT-DE) [Horndeski, 1974, Gubitosi et al., 2013, Bellini and Sawicki, 2015] describes the most general scalar-tensor theory in the unitary gauge via the action

$$S = \int d^4x\,\sqrt{-g}\,\Big[\frac{M_{\rm Pl}^2}{2} f(t) R - \Lambda(t) - c(t)\,g^{00} + \frac{M_2^4(t)}{2}\,(\delta g^{00})^2 - \frac{\bar{M}_1^3(t)}{2}\,\delta K\,\delta g^{00} - \frac{\bar{M}_2^2(t)}{2}\,\delta K^2 + \cdots\Big], \tag{42}$$

where $\delta K = K - 3H$ is the perturbation of the extrinsic curvature trace. The $\bar{M}_2^2$ operator (extrinsic-curvature-squared term; related to the running Planck mass α_M in the Bellini-Sawicki basis, distinct from the braiding operator $\bar{M}_1^3$) contributes an effective sound speed $c_s^2 \to c_s^2 + \bar{M}_2^2/(M_{\rm Pl}^2)$ at linear order.

6.2 Mapping IS viscosity to EFT operators

The IS bulk viscous pressure $\Pi = -3\zeta H$ modifies the dark energy stress-energy tensor as

$$T_{\mu\nu}^{\rm DE} \to T_{\mu\nu}^{\rm DE} + \Pi\,(g_{\mu\nu} + u_\mu u_\nu), \tag{43}$$

where Π acts as an additional isotropic pressure. At the perturbation level, the bulk viscous pressure $\delta\Pi$ generates a term in the DE pressure perturbation: $\delta p_{\rm DE} = c_s^2\,\delta\rho_{\rm DE} + \delta\Pi$.

Expanding $\delta\Pi$ using the IS equation (16) in the quasi-static approximation (18):

$$\delta\Pi_{\rm QS} = -\tilde{\zeta}\,\rho_{\rm DE}\,\frac{k}{aH}\,v_{\rm DE} = -\frac{\tilde{\zeta}\,\rho_{\rm DE}}{aH}\,(\mathrm{div}\,\mathbf{v}), \tag{44}$$

which is proportional to the DE velocity divergence — a structure analogous to the $\bar{M}_2^2$ operator in the EFT-DE action (note: true kinetic braiding corresponds to $\bar{M}_1^3\delta K\delta g^{00}$ in the Bellini-Sawicki basis [Bellini and Sawicki, 2015]; the mapping here is at the level of the effective DE pressure perturbation, not operator identification). The dimensional mapping is:

$$\bar{M}_2^2/M_{\rm Pl}^2 = \frac{\tilde{\zeta}}{\tau_\Pi H_0} = \Omega_{\rm DE}, \tag{45}$$

where we have used the SSEE identity (11). Equation (45) identifies the IS viscosity with a specific EFT operator combination: the EFT coefficient is exactly $\Omega_{\rm DE}$, a pure algebraic prediction of SSEE.

6.3 Comparison with ghost condensate

The ghost condensate [Arkani-Hamed et al., 2004] achieves gradient stability via the EFT operator $M_{\rm Pl}^2\,\alpha_K\,(\delta g^{00})^2/2$, which generates $c_s^2 = 1$ for the scalar perturbation — the opposite extreme from SSEE. Table 6 summarises the key differences.

Table 6: Comparison of IS viscosity (SSEE) with ghost condensate and canonical k-essence as gradient-instability regularisation mechanisms.

Property	SSEE IS	Ghost condensate	k-essence
$c_{s,\rm eff}^2$	0 (exact)	1	$w_0 < 0$
Free parameters	0	$M_{\rm Pl}^2\alpha_K$	n
Causal speed $v_{\rm sig}/c$	1 (saturates)	< 1	N/A
EFT operator	$\bar{M}_2^2 \propto \Omega_{\rm DE}$	$M_{\rm Pl}^2\alpha_K$	$c_s^2 g^{00}$
DE clustering	$\delta_{\rm DE} \to 0$	$\delta_{\rm DE} \sim \delta_m$	unstable
S_8 prediction	0.725 ± 0.005	~ 0.811	(unstable)
DESI DR2 χ^2	0.05σ	$> 3\sigma$	$> 3\sigma$

The key distinction is that SSEE IS achieves *exact marginal stability* ($c_s^2 = 0$), which is the only value consistent simultaneously with causal propagation at speed c (Section 2.3) and zero free parameters. Ghost condensation with $c_s^2 = 1$ predicts DE clustering at the level of matter clustering ($\delta_{\rm DE} \sim \delta_m$), generating a large effective S_8 inconsistent with current weak-lensing data.

7 Discussion

7.1 Physical origin of the MIRA factor

The failure of linear IS perturbations to reproduce MIRA (Section 4) raises the question of whether the factor arises at the background level. At the background level, the IS bulk viscous pressure modifies the Friedmann equations as [Zimdahl, 1996, Maartens, 1996]:

$$3H^2 M_{\rm Pl}^2 = \rho_m + \rho_{\rm DE} + \Pi_{\rm bulk}, \tag{46}$$

where in the SSEE steady state $\Pi_{\rm bulk} = -3\zeta H = -\mathrm{KAL}_0\,\rho_{\rm DE} H$. This viscous term effectively shifts the observed matter density. However, numerical integration of the background IS transport equations reveals that fitting an effective Ω_m to the resulting $H(z)$ evolution yields $\Omega_{m,\rm fit}/\Omega_{m,\rm dyn} \approx 3.04$ when using the SSEE viscous constant $\zeta = \mathrm{KAL}_0/3$.

This is 52% higher than the algebraic MIRA factor (1.9989). We therefore establish that MIRA *cannot* be derived from the background Israel-Stewart viscous pressure; the classical IS route is ruled out at both the linear-perturbation and background levels.

Paper 6 of this series provides the physical basis: the MIRA identity reflects a *two-sector* dark matter structure in which a ϕ-dark matter component with $\Omega_{\phi\text{-DM}} = (\mathrm{MIRA}-1)\,\Omega_{m,\rm dyn} = 0.160$ and algebraically fixed mass $m_\phi = 5.71\,\mathrm{eV}$ contributes at scales below the free-streaming cutoff $k_{\rm fs} = 0.493\,h\,\mathrm{Mpc}^{-1}$. The near-equality $\Omega_{\phi\text{-DM}} \approx \Omega_{m,\rm dyn}$ produces MIRA ≈ 2; the precise value $(3\varphi+\pi)/4$ traces back to the algebraic mass constraint $m_\phi = \Sigma m_\nu^{\rm active} \times H_{0,\rm alg}$. This phenomenological resolution is subject to the Ly-α and KATRIN/PTOLEMY falsification criteria described in Paper 6; a first-principles derivation from quantum field theory remains an open problem.

7.2 Implications for JWST early galaxy counts

The exact marginal stability $c_{s,\rm eff}^2 = 0$ has a specific prediction for structure formation: DE perturbations at sub-horizon scales effectively vanish, leaving matter to cluster in a background with effective matter density $\Omega_{m,\rm dyn} = 0.160$. The lower matter density delays structure formation relative to ΛCDM in one sense (less matter to cluster), but the reduced Jeans mass from the pressureless DE sector ($c_s^2 = 0$) and the modified collapse threshold $\delta_c^{\rm SSEE} = 1.6284$ [Almeida Vallejo, 2026d] (compared to the EdS value 1.686) boost the halo mass function at high mass and high redshift.

Using the Press-Schechter formalism with the SSEE collapse threshold [Almeida Vallejo, 2026d], the excess of massive halos ($M > 10^{11}\,M_\odot$) at $z > 10$ relative to ΛCDM is

$$\frac{n_{\rm SSEE}(M,z)}{n_{\Lambda\rm CDM}(M,z)} \approx 1.4\text{–}2.0 \quad (M > 10^{11}\,M_\odot,\ z > 10), \tag{47}$$

potentially explaining the excess of bright galaxies reported by Boylan-Kolchin [2023] in JWST observations at $z > 10$. This prediction will be quantitatively testable when JWST halo mass functions at $z \gtrsim 12$ become available at the $\lesssim 20\%$ level.

7.3 The S_8–$f\sigma_8$ split and the path to new physics

The numerical results expose a structural split between two growth observables that probe complementary aspects of structure formation:

- S_8 **(weak-lensing amplitude):** SSEE predicts $S_8 = 0.725 \pm 0.005$, within 1.96σ of KiDS-1000 and 2.84σ of DES-Y3. Both tensions are *smaller* than those of ΛCDM (3.27σ and 3.26σ respectively), achieved without any free parameter adjustment.
- $f\sigma_8$ **(RSD growth rate):** SSEE predicts $f\sigma_8(z = 0.5) = 0.341$, below the BOSS DR12 value of 0.458 ± 0.038 by 3.1σ. The mean tension across six RSD surveys is 2.67σ, compared to 0.73σ for ΛCDM.

The two observables are not contradictory: $S_8 = \sigma_8\sqrt{\Omega_m/0.3}$ weights the growth *amplitude* σ_8, while $f\sigma_8 = f(z)\,\sigma_8\,D_1(z)/D_1(0)$ also carries the instantaneous growth *rate* $f(z) \approx \Omega_m(z)^\gamma$. In SSEE, $\Omega_m(z)$ is small because $\Omega_{m,\rm dyn} = 0.160$, suppressing $f(z)$ at all redshifts even though the integrated growth amplitude σ_8 is well-matched to weak lensing.

This split is a diagnostic, not a refutation. It identifies precisely where the current SSEE framework is incomplete: the *effective gravitational coupling* for matter perturbations is set by $\Omega_{m,\rm dyn}$, whereas observations require an effective coupling closer to $\Omega_{m,\rm CMB} \simeq 0.315$. Within the Israel-Stewart EFT framework of Paper 1 [Almeida Vallejo, 2026a], the natural extension is a non-minimal *kinetic braiding* coupling (the α_B parameter in the Bellini-Sawicki basis [Bellini and Sawicki, 2015]), which can enhance the effective growth source $G_{\rm eff}/G_N = 1 + \alpha_B\,\Omega_{\rm DE}/\Omega_m$ without altering $\Omega_{m,\rm dyn}$, the background $H(z)$, or σ_8. Whether SSEE forces a specific algebraic value of α_B from φ and π is an open question reserved for future work.

7.4 The H_0 tension and SSEE

SSEE predicts $H_0 = 3(\varphi+\pi)^2 \simeq 67.96\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ [Almeida Vallejo, 2026d], consistent with Planck 2018 (67.36 ± 0.54) at 1.1σ and with DESI DR2. The IS mechanism does not modify H_0 directly, since the viscous pressure $\Pi_{\rm bulk}$ averages to zero at early times ($z > 5$) when $\Omega_{\rm DE} \to 0$. The H_0 tension between Planck and SH0ES therefore remains at the same level in SSEE as in ΛCDM — SSEE does not worsen nor resolve the H_0 tension, but introduces a new prediction for σ_8 and S_8 that partially resolves the lensing tension.

7.5 Causal structure and Hiscock-Lindblom stability

The IS theory satisfies the Hiscock-Lindblom stability criteria [Hiscock and Lindblom, 1983]: positivity of entropy production and causal propagation, provided $\tau_\Pi > 0$, $\zeta \geq 0$. We have shown that SSEE satisfies these conditions (§2.3). Moreover, the signal speed $v_{\rm sig} = c$ saturates the causality bound, meaning SSEE IS dark energy is maximally causal: perturbations propagate at the speed of light, with no superluminal modes. This distinguishes SSEE from first-order (Eckart) viscous theories, which are acausal [Hiscock and Lindblom, 1985] and would introduce spurious instabilities in the CMB power spectrum at $\ell > 1000$.

7.6 Falsifiable predictions summary

8 Conclusions

We have performed a linear Israel-Stewart causal viscous perturbation analysis in the sub-horizon regime analysis of SSEE dark energy, addressing three physically motivated questions.

Q1 — Gradient stability (proved analytically). The SSEE structure constants satisfy the exact algebraic identity $\tilde{\zeta}/(\tau_\Pi H_0) = \Omega_{\rm DE} = |w_0|$, which forces $c^2_{s,\rm eff}(k \to \infty) = w_0 + |w_0| = 0$ to machine precision. This is a theorem — not a numerical coincidence or parameter tuning. The critical wavenumber $k_{\rm crit} \simeq 0.456\,H_0/c < H_0/c$ lies outside the sub-horizon regime; all observationally relevant modes are causally stabilised. Moreover,

Table 7: Falsifiable predictions of SSEE IS dark energy, with observational targets and current/future datasets capable of testing each prediction. A single failure at the stated level would falsify SSEE-V3.6.

Prediction	SSEE value	Test level	Dataset / Timeline
$c^2_{s,\mathrm{eff}} = 0$ (DE pressureless)	0 (exact)	$\lvert c_s^2\rvert < 0.05$	CMB lensing/$T_{\ell\ell'}$ cross [Calabrese et al., 2011] / current
$S_8 = 0.725$	0.725 ± 0.005	$< 3\sigma$ of KiDS, DES	KiDS-1000, DES-Y3 / current
$\gamma_{\mathrm{IS}} = 0.554$	0.554 ± 0.001	test via $f\sigma_8$; γ degenerate	DESI full shape, Euclid / 2026–2028
$f\sigma_8(z = 0.5) = 0.341$	0.341 ± 0.010	28% below ΛCDM; current RSD tension 2.67σ (6 surveys)	DESI full-shape / 2026
D_1 suppression $\mathcal{G} = 0.866$	0.866 ± 0.005	2% precision	Euclid weak lensing / 2027
No DE clustering ($\lvert\delta_{\mathrm{DE}}/\delta_m\rvert < 10^{-3}$)	$< 10^{-3}$ at $k > 20\,H_0/c$	cosmic variance	Future 21cm surveys / 2030+
MIRA $= (3\varphi + \pi)/4$ (two-sector ϕ-DM basis, Paper 6)	1.9989	Ω_m direct measurement	Stage IV CMB / 2030

the signal speed saturates the causality bound ($v_{\mathrm{sig}} = c$), placing SSEE at the maximally causal, minimally fine-tuned configuration consistent with thermodynamic stability.

Q2 — MIRA factor (pure algebraic axiom). Linear IS perturbations yield $\mathrm{MIRA}_{\mathrm{num}} = 0.989 \pm 0.017$ versus $\mathrm{MIRA}_{\mathrm{alg}} = 1.9989$ at all sub-horizon scales. The discrepancy is 50%, establishing that the MIRA factor is not a perturbative phenomenon. Furthermore, numerical integration of the background IS Friedmann equations yields an effective ratio of 3.04, failing to reproduce MIRA at the background level. We conclude that the MIRA factor does not emerge from classical Israel-Stewart fluid mechanics. Paper 6 proposes that the physical basis is instead a two-sector dark matter structure (CDM + ϕ-DM with $m_\phi = 5.71\,\mathrm{eV}$), whose near-equal densities produce MIRA ≈ 2 algebraically; a first-principles derivation from quantum field theory remains open.

Q3 — Growth rate and S_8 (partial resolution of lensing tension). The IS growth index $\gamma_{\mathrm{IS}} = 0.554 \pm 0.001$ is numerically indistinguishable from $\gamma_{\Lambda\mathrm{CDM}} \simeq 0.55$ — the power-law exponent $f \approx \Omega_m(a)^\gamma$ is insensitive to the background model. However, the *growth rate itself* is suppressed because $\Omega_{m,\mathrm{dyn}} = 0.160$: the linear suppression is $\mathcal{G} = 0.866$, giving $\sigma_8^{\mathrm{SSEE}} = 0.702 \pm 0.005$ and $S_8^{\mathrm{SSEE}} = 0.725 \pm 0.005$. This leaves SSEE in 2.0σ and 2.8σ tension with KiDS-1000 and DES-Y3 respectively. The IS growth sector is explored at the linear sub-horizon level considered in this paper; the two-sector ϕ-DM construction of Paper 6 provides the physical basis for MIRA. The lowered amplitude arises from a structural consequence of IS dynamics with $\Omega_{m,\mathrm{dyn}} = 1 + w_0 = 1 - T_r/M_v \simeq 0.160$. By contrast, the growth *rate* $f(z) \approx \Omega_m(z)^\gamma$ is suppressed at all redshifts by the same low $\Omega_{m,\mathrm{dyn}}$, producing a mean $f\sigma_8$ tension of 2.67σ across six RSD surveys — a challenge that cannot be resolved by parameter tuning and is reported transparently as a diagnostic of the sector where the current framework requires extension (Section 7.3).

Together, these results establish that SSEE dark energy is: (i) causally and thermodynamically stable (Q1), (ii) predictive of an S_8 value within 2.0σ and 2.8σ of KiDS and DES respectively, exploring the growth tension without free parameter adjustment (Q3), and (iii) strictly bounded: the MIRA factor cannot be derived from IS dynamics (Q2); the physical basis is provided by Paper 6's two-sector ϕ-dark matter construction ($\Omega_{\phi\text{-DM}} \approx \Omega_{m,\mathrm{dyn}}$ gives MIRA≈ 2; exact value from the algebraic mass $m_\phi = 5.71\,\mathrm{eV}$), with a first-principles QFT derivation remaining open. The primary falsifiable prediction is $f\sigma_8(z = 0.5) = 0.341$, which is 28% below ΛCDM. Comparison with current RSD data (6dFGRS, SDSS MGS, BOSS DR12, eBOSS DR16; Table 5) yields a mean tension of 2.67σ across six surveys. Like the $H(z)$ chronometer tension, this $f\sigma_8$ suppression is a direct physical consequence of the low geometric matter density ($\Omega_{m,\mathrm{dyn}} = 0.160$). It functions as a diagnostic suggesting that a future theoretical extension, for example via kinetic braiding in an EFT framework, may be required to dynamically enhance the effective gravitational clustering force without altering the underlying geometric axioms. This tension is reported transparently; a confirmed $> 5\sigma$ discrepancy from DESI Year-1 full-shape analysis would directly falsify the current unbraided SSEE-V3.6 formulation.

Data Availability

The Python script `src/ssee_paper5_IS_perturbations.py` (including the growth rate and S_8 computations) and all output figures are available in the SSEE-V3.6 repository at `https://github.com/mikealmeida1721/SSEE` (Zenodo DOI: 10.5281/zenodo.19679049). All computations use publicly available Python packages (NumPy, SciPy, Matplotlib).

Acknowledgements

This research was carried out as an independent investigation. The author thanks the open-source communities behind NumPy, SciPy, Matplotlib, and CAMB, and acknowledges the Planck, DESI, KiDS, and DES teams for making their data publicly available.

A Derivation of the IS Dispersion Relation

We derive Eq. (21) from the linearised IS equations in Minkowski space (the cosmological curvature terms are subdominant in the sub-horizon limit).

Linearised IS system. Write $\rho = \bar\rho + \delta\rho$, $p = \bar p + \delta p$, $\Pi = \bar\Pi + \delta\Pi$, $u^\mu = (1, \mathbf{v})$ with $|\mathbf{v}| \ll 1$. The continuity and Euler equations give:

$$\dot{\delta\rho} + (\bar\rho + \bar p)\,\nabla\cdot\mathbf{v} = 0, \tag{48}$$

$$(\bar\rho + \bar p)\,\dot{\mathbf{v}} + \nabla(\delta p + \delta\Pi) = 0. \tag{49}$$

The IS relaxation equation linearises to:

$$\tau_\Pi \, \delta\dot{\Pi} + \delta\Pi = -\zeta \, \nabla \cdot \mathbf{v}. \tag{50}$$

Plane-wave ansatz. For modes $e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}$:

$$\delta\Pi = -\frac{i\zeta k}{\tau_\Pi(-i\omega)+1}\, v_\parallel, \tag{51}$$

where $v_\parallel = \mathbf{k}\cdot\mathbf{v}/k$ is the longitudinal velocity.

Effective sound speed. Combining with the Euler equation:

$$\omega^2 = \left[c_{s,\mathrm{bare}}^2 + \frac{(\zeta/\tau_\Pi\rho)}{1+(\omega\tau_\Pi)^{-2}}\right] k^2. \tag{52}$$

In the IS limit $|\omega|\tau_\Pi \gg 1$ (short wavelengths):

$$\omega^2 \approx \left(c_{s,\mathrm{bare}}^2 + \frac{\zeta}{\tau_\Pi\,\rho}\right) k^2 = c_{s,\mathrm{eff}}^2\, k^2. \tag{53}$$

Using the SSEE normalisation $\zeta = \tilde{\zeta}\,\rho_{\mathrm{DE}} H_0$ and $\rho = \rho_{\mathrm{DE}}$:

$$c_{s,\mathrm{eff}}^2 = c_{s,\mathrm{bare}}^2 + \frac{\tilde{\zeta} H_0}{\tau_\Pi} = c_{s,\mathrm{bare}}^2 + \frac{\tilde{\zeta}}{\tau_\Pi H_0}, \tag{54}$$

recovering Eq. (21). The full k-dependent interpolation $c_{s,\mathrm{eff}}^2(k) = c_{s,\mathrm{bare}}^2 + (\tilde{\zeta}/\tau_\Pi H_0)\, x^2/(1+x^2)$ follows from retaining the full ω dependence.

B Eigenvalue Analysis of the IS Perturbation Matrix

The quasi-static 4-variable system $\mathbf{Y} = [\delta_m, v_m, \delta_{\mathrm{DE}}, v_{\mathrm{DE}}]^\top$ has Jacobian matrix $\mathbf{J}$ (evaluated at $z = 0$, $k = 10\,H_0/c$):

$$\mathbf{J} = \begin{pmatrix} 0 & -k/H & 0 & 0 \\ -3\Omega_m/2k & -1 & -3\Omega_{\mathrm{DE}}/2k & 0 \\ 0 & 0 & 0 & -(1+w)k/H \\ -3\Omega_m/(2k(1+w)) & 0 & -c_s^2 k/((1+w)H) & -\mathcal{F} \end{pmatrix}, \tag{55}$$

where $\mathcal{F} = (1-3c_s^2) + \tilde{\zeta}(k/H)^2 \approx 186$ at $k = 10$, $z = 0$.

The eigenvalues λ_i (computed numerically from $\det(\mathbf{J}-\lambda\mathbf{I}) = 0$):

$$\lambda_1 \approx +0.15 \quad \text{(growing matter mode)}, \tag{56}$$
$$\lambda_2 \approx -1.15 \quad \text{(decaying matter mode)}, \tag{57}$$
$$\lambda_3 \approx +0.02 \quad \text{(IS-damped DE, slowly growing)}, \tag{58}$$
$$\lambda_4 \approx -\mathcal{F} \approx -186 \quad \text{(fast IS relaxation)}. \tag{59}$$

The large negative eigenvalue $\lambda_4 \approx -186$ drives v_{DE} to its quasi-static value on a timescale $\sim (186\,H)^{-1}$, justifying the quasi-static approximation. The small positive $\lambda_3 \approx +0.02$ means DE perturbations grow very slowly compared to matter ($\lambda_1 \approx 0.15$), consistent with $|\delta_{\mathrm{DE}}/\delta_m| \ll 1$ at $z = 0$. No eigenvalue has a growth rate exceeding λ_1: the system is perturbatively stable with the matter growing mode dominating.

References

M. S. Abdul-Karim et al. DESI 2025 DR2: Baryon acoustic oscillations from the complete survey. *arXiv*, 2025.

S. Alam et al. The clustering of galaxies in the completed SDSS-III baryon oscillation spectroscopic survey: cosmological analysis of the DR12 galaxy sample. *Monthly Notices of the Royal Astronomical Society*, 470:2617–2652, 2017. doi: 10.1093/mnras/stx721.

Mike Edison Almeida Vallejo. A zero-parameter framework for cosmological dynamics and galaxy cluster mass discrepancies via structural self-energy expansion (SSEE-V3.6). *arXiv*, 2026a. SSEE Paper 1.

Mike Edison Almeida Vallejo. Bayesian validation of the SSEE-V3.6 framework against DESI DR2, Planck 2018, and galaxy cluster data. *arXiv*, 2026b. SSEE Paper 2.

Mike Edison Almeida Vallejo. CMB planck PR4 confrontation of the SSEE-V3.6 dark energy framework. *arXiv*, 2026c. SSEE Paper 3.

Mike Edison Almeida Vallejo. Algebraic derivation of CMB parameters from φ and π: The nine sovereignties of SSEE-V3.6. *arXiv*, 2026d. SSEE Paper 4.

N. Arkani-Hamed, H.-C. Cheng, M. A. Luty, and S. Mukohyama. Ghost condensation and a consistent infrared modification of gravity. *Journal of High Energy Physics*, 2004: 074, 2004. doi: 10.1088/1126-6708/2004/05/074.

C. Armendariz-Picon, V. Mukhanov, and P. J. Steinhardt. Essentials of k-essence. *Physical Review D*, 63:103510, 2001. doi: 10.1103/PhysRevD.63.103510.

M. Asgari et al. KiDS-1000 cosmology: Cosmic shear constraints and comparison between two point statistics. *Astronomy & Astrophysics*, 645:A104, 2021. doi: 10.1051/0004-6361/202039070.

E. Bellini and I. Sawicki. Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity. *Journal of Cosmology and Astroparticle Physics*, 2015(07):050, 2015. doi: 10.1088/1475-7516/2015/07/050.

F. Beutler et al. The 6df galaxy survey: baryon acoustic oscillations and the local hubble constant. *Monthly Notices of the Royal Astronomical Society*, 423:3430–3444, 2012. doi: 10.1111/j.1365-2966.2012.21136.x.

M. Boylan-Kolchin. Stress testing ΛCDM with high-redshift galaxy candidates. *Nature Astronomy*, 7:728–735, 2023. doi: 10.1038/s41550-023-01937-7.

I. Brevik and Ø. Grøn. Universe models with negative bulk viscosity. *Astrophysics and Space Science*, 298:315–325, 2005. doi: 10.1007/s10509-005-5234-7.

E. Calabrese, D. N. Spergel, B. D. Sherwin, and R. Nagata. Non-adiabatic dark energy perturbations. *Physical Review D*, 83:023011, 2011. doi: 10.1103/PhysRevD.83.023011.

R. R. Caldwell. A phantom menace? cosmological consequences of a dark energy component with super-negative equation of state. *Physics Letters B*, 545:23–29, 2002. doi: 10.1016/S0370-2693(02)02589-3.

M. Chevallier and D. Polarski. Accelerating universes with scaling dark matter. *International Journal of Modern Physics D*, 10:213–224, 2001. doi: 10.1142/S0218271801000822.

Dark Energy Survey Collaboration. Dark Energy Survey year 3 results: Cosmological constraints from galaxy clustering and weak lensing. *Physical Review D*, 105:023520, 2022. doi: 10.1103/PhysRevD.105.023520.

G. S. Denicol, H. Niemi, E. Molnar, and D. H. Rischke. Derivation of transient relativistic fluid dynamics from the Boltzmann equation. *Physical Review D*, 85:114047, 2012. doi: 10.1103/PhysRevD.85.114047.

DESI Collaboration, A. G. Adame, et al. DESI 2024 V: Full-Shape Galaxy Clustering from Galaxies and Quasars. *arXiv e-prints*, 2024.

C. Eckart. The thermodynamics of irreversible processes. III. relativistic theory of the simple fluid. *Physical Review*, 58:919–924, 1940. doi: 10.1103/PhysRev.58.919.

G. Gubitosi, F. Piazza, and F. Vernizzi. The effective field theory of dark energy. *Journal of Cosmology and Astroparticle Physics*, 2013(02):032, 2013. doi: 10.1088/1475-7516/2013/02/032.

W. A. Hiscock and L. Lindblom. Stability and causality in dissipative relativistic fluids. *Annals of Physics*, 151:466–496, 1983. doi: 10.1016/0003-4916(83)90288-9.

W. A. Hiscock and L. Lindblom. Generic instabilities in first-order dissipative relativistic fluid theories. *Physical Review D*, 31:725–733, 1985. doi: 10.1103/PhysRevD.31.725.

G. W. Horndeski. Second-order scalar-tensor field equations in a four-dimensional space. *International Journal of Theoretical Physics*, 10:363–384, 1974. doi: 10.1007/BF01807638.

J. Hou et al. eBOSS collaboration: The completed SDSS-IV extended baryon oscillation spectroscopic survey: BAO and RSD measurements from anisotropic clustering analysis of the quasar sample between redshift 0.8 and 2.2. *Monthly Notices of the Royal Astronomical Society*, 500:1201–1221, 2021. doi: 10.1093/mnras/staa3234.

C. Howlett, A. J. Ross, L. Samushia, W. J. Percival, and M. Manera. The clustering of the SDSS main galaxy sample – II. mock galaxy catalogues and a measurement of the growth of structure from redshift space distortions at $z = 0.15$. *Monthly Notices of the Royal Astronomical Society*, 449:848–866, 2015. doi: 10.1093/mnras/stu2693.

W. Hu. Structure formation with generalized dark matter. *The Astrophysical Journal*, 506:485–494, 1998. doi: 10.1086/306274.

W. Israel. Nonstationary irreversible thermodynamics: A causal relativistic theory. *Annals of Physics*, 100:310–338, 1976. doi: 10.1016/0003-4916(76)90064-6.

W. Israel and J. M. Stewart. Transient relativistic thermodynamics and kinetic theory. *Annals of Physics*, 118:341–372, 1979. doi: 10.1016/0003-4916(79)90130-1.

O. Lahav, P. B. Lilje, J. R. Primack, and M. J. Rees. Dynamical effects of the cosmological constant. *Monthly Notices of the Royal Astronomical Society*, 251:128–136, 1991. doi: 10.1093/mnras/251.1.128.

E. V. Linder. Exploring the expansion history of the universe. *Physical Review Letters*, 90:091301, 2003. doi: 10.1103/PhysRevLett.90.091301.

E. V. Linder. Cosmic growth history and expansion history. *Physical Review D*, 72: 043529, 2005. doi: 10.1103/PhysRevD.72.043529.

R. Maartens. Causal thermodynamics in relativity. *arXiv*, 1996.

I. Müller. Zum paradoxon der wärmeleitungstheorie. *Zeitschrift für Physik*, 198:329–344, 1967. doi: 10.1007/BF01326412.

Planck Collaboration. Planck 2018 results. VI. cosmological parameters. *Astronomy & Astrophysics*, 641:A6, 2020. doi: 10.1051/0004-6361/201833910.

D. H. Weinberg, M. J. Mortonson, D. J. Eisenstein, C. Hirata, A. G. Riess, and E. Rozo. Observational probes of cosmic acceleration. *Physics Reports*, 530:87–255, 2013. doi: 10.1016/j.physrep.2013.05.001.

S. Weinberg. Entropy generation and the survival of protogalaxies in an expanding universe. *The Astrophysical Journal*, 168:175–194, 1971. doi: 10.1086/151073.

W. Zimdahl. Bulk viscous cosmology. *Physical Review D*, 53:5483–5493, 1996. doi: 10.1103/PhysRevD.53.5483.